

AI-Powered Fire Spread Prediction and Containment System

System Requirements Specification

Panic! At The Kernel
Client: EPI-USE Africa

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1 Introduction

AI-Powered Fire Spread Prediction and Containment System, is designed to address the critical business need for real-time situational awareness and rapid emergency response by utilizing AI-driven simulations and multi-source verification to mitigate the impact of fire disasters. Developed according to IEEE 29148 and ISO 22320 standards, the project's scope encompasses an end-to-end ecosystem that includes secure authentication and role-based access control for users, firefighters, and administrators. The core functionality features a public fire reporting interface with automated verification which triggers proximity-based push notifications for registered users. Furthermore, the system includes a live mapping interface, specialized tactical tools for firefighters to log containment lines and view AI fire spread simulation, and an offline map cache to ensure operational continuity in areas with limited connectivity.

2 User Characteristics and User Stories

2.1 User Characteristics

The system supports four user classes, each with distinct access levels and responsibilities.

- **Anonymous Visitor:** Can report fires and view the live map without an account.
- **Registered User:** Can report fires and receive push notifications.
- **Firefighter:** Has access to tactical tools, containment line logging, and AI fire spread simulation.
- **Administrator:** Manages user roles, reviews fire reports, and maintains the incident archive.

2.2 User Stories

2.2.1 E1: Authentication and Access Control

- As an anonymous visitor, I want to register with an email and password so that I can receive push alerts.
- As a registered user, I want to log in so that I can access my role-based dashboard.
- As an administrator, I want to approve or revoke user roles so that only authorised personnel can access operational tools.
- As a firefighter, I want to log in so that I can access my role-based dashboard, firefighter tactical tools, and AI simulation.

2.2.2 E2: Fire Reporting and Verification

- As any user, I want to report a fire without logging in so that I can alert others immediately.
- As the system, I want to auto-verify reports so that false alerts are minimised.
- As a firefighter or administrator, I want to manually verify or reject pending reports so that unconfirmed incidents receive appropriate review.

2.2.3 E3: AI Fire-Spread Simulation

- As the system, I want to generate a fire-spread prediction on specific verified fire report.
- As the system, I want to re-run the simulation when a containment line is logged so that the prediction reflects current field actions.
- As the system, I want to display time lapsed simulation when prompted.

2.2.4 E4: Live Fire Map

- As any user, I want to view the live fire map without logging in so that I can see active fires near me.

2.2.5 E5: Notification and Alert System

- As a registered user, I want to receive a push notification when a fire is verified near me so that I can take immediate action.

2.2.6 E6: Firefighter Tactical Tools

- As a firefighter, I want a tactical dashboard with AI recommendations so that I can make informed decisions in the field.
- As a firefighter, I want to draw containment lines on the map so that the AI simulation reflects my ground actions.

2.2.7 E7: Offline Map Cache

- As a firefighter, I want to view cached map data when offline so that I retain situational awareness without connectivity.

2.2.8 E8: Administration Panel

- As an administrator, I want a management panel to approve role requests and revoke access so that system permissions remain controlled.

3 Functional Requirements

3.1 E1: Authentication and Access Control

- R1.1 The home screen shall present three options: Log In, Register, and Login as Guest.
- R1.2 New registrations shall automatically be assigned the Registered User role.
- R1.3 The Administrator role shall require approval by an existing Administrator. The first administrator shall be provisioned by the system seed at deployment time.
- R1.4 Password reset shall be performed via a time-limited email link. The link shall expire after 15 minutes and be single-use.
- R1.5 All role and permission checks shall be enforced server-side at the API layer.
- R1.6 JWT tokens shall be used for all authenticated API calls. Access tokens shall be short-lived (e.g., 15 minutes) and paired with refresh tokens.
- R1.7 All client-server communication shall use HTTPS/TLS.
- R1.8 All authentication events and role changes shall be written to the audit log.
- R1.9 Tokens shall be invalidated immediately on logout, role change, or account suspension. Because pure JWTs cannot be instantly revoked without a denylist, the system relies on short TTL and refresh token rotation.
- R1.10 Two-factor authentication (TOTP) shall be available for all user roles. Users may enable 2FA after login; a setup endpoint returns an otpauth:// URI.

R1.11 The system shall enforce rate limiting on login attempts. Limit: 5 failed attempts per 15 minutes per account. After 10 consecutive failures, the account shall be locked for 30 minutes.

3.2 E2: Fire Reporting and Verification

- R2.1 An unregistered user shall be able to submit a fire report.
- R2.2 The fire reporting map shall be accessible from the home screen without login.
- R2.3 Required report inputs shall be: map pin location, estimated fire size and photo attachment.
- R2.4 Optional report fields shall include: free-text description.
- R2.5 The system shall generate a unique reference number for every report.
- R2.6 All reports shall enter Pending status; no alerts shall be triggered until Verified.
- R2.7 GPS auto-fill shall be offered; manual map pin placement shall always be available.
- R2.8 Duplicate reports at the same location shall be automatically linked to one incident.
- R2.9 Auto-verification shall trigger on two independent same-location reports.
- R2.10 Manual verification and rejection shall be available to Firefighter and Administrator roles.
- R2.11 Rejection shall require a mandatory reason field.
- R2.12 Registered reporters shall be notified on verification or rejection of their report.
- R2.13 All reports shall be stored in the database for auditing purposes.
- R2.14 Report status values shall be: Pending, Verified, and Rejected.

3.3 E3: AI Fire-Spread Simulation

- R3.1 The simulation engine shall ingest: wind speed and direction, temperature, humidity, precipitation, and vegetation dryness index.
- R3.2 The simulation engine shall account for terrain slope and vegetation type.
- R3.3 Output shall be a spread probability grid for 1 h, 3 h, 24 h, and 72 h horizons. Grid resolution: 10 m × 10 m cells. Probability scale: 5 discrete bands (0–20%, 21–40%, 41–60%, 61–80%, 81–100%).
- R3.4 The initial prediction shall be delivered within 5 seconds of verification under normal load (one active fire, weather cache warm). If the simulation exceeds 10 seconds, the engine shall fall back to the last valid prediction (if any) and mark it as “degraded”; otherwise, it shall return a wind-biased elliptical buffer around the ignition point.
- R3.5 The engine shall when a containment line is logged.
- R3.6 The simulation engine shall be versioned. Each prediction shall record the model version (e.g., “wind.ca.v1.0”). Model version and input timestamps shall be stored in the audit log to enable post-hoc reproducibility.
- R3.7 The baseline implementation shall be a wind-biased cellular automaton (CA) model. Fire spreads from ignition boundary radius to neighbouring cells with probabilities proportional to wind speed and direction, reduced by humidity and vegetation dryness. A convolutional neural network (CNN) with LSTM layers for temporal weather trends is a stretch goal, not required for the first demonstrator.

- R3.8 The system shall meet a minimum accuracy target evaluated against historical fire incidents. For a held-out test set of at least 15 historical fires (ignition point, weather, and observed final perimeter), the 6-hour predicted spread polygon shall achieve an Intersection-over-Union (IoU) of at least 0.20 against the observed burned area. This metric shall be recomputed after each major model update.

3.4 E4: Live Fire Map

- R4.1 The map shall load without login; verified fire markers shall be visible to all users.
- R4.2 Tapping a fire marker shall display: status, report time, and estimated fire size.
- R4.3 The map shall auto-refresh on new data with no manual reload required.
- R4.4 The map shall be responsive on mobile, tablet, and desktop.
- R4.5 Map tiles shall be cached locally to support basic offline viewing.

3.5 E5: Notification and Alert System

- R5.1 Push alerts shall be dispatched within seconds of a fire report reaching Verified status.
- R5.2 Alerts shall be delivered as push notifications to registered users.
- R5.3 Notifications shall appear when the app is closed or running in the background.
- R5.4 Topic-based grid-zone messaging shall be used for simultaneous large-scale delivery.
- R5.5 The message broker shall retain and retry failed deliveries.
- R5.6 Notification content shall include: fire location, distance from user, severity, and a map deep-link.
- R5.7 Every dispatched alert shall be logged with a timestamp, zone ID, and distance.

3.6 E6: Firefighter Tactical Tools

- R6.1 Firefighters shall have access to a tactical dashboard displaying a live fire map.
- R6.2 The dashboard shall show: fire direction, high-risk areas, wind speed and direction, and suggested containment locations.
- R6.3 Firefighters shall be able to draw containment lines directly on the map.
- R6.4 Every logged action shall be stored with GPS coordinates and a timestamp.
- R6.5 The interface shall be optimised for field use with large tap targets and high contrast.

3.7 E7: Offline Map Cache

- R7.1 The app shall cache map tiles and the most recent spread prediction while online.
- R7.2 An offline indicator shall be displayed when the device has no connectivity.
- R7.3 Cached map data and the most recent prediction shall remain viewable while offline.

3.8 E8: Administration Panel

R8.1 The admin panel shall support: approving and rejecting role requests, searching users, and revoking access.

R8.2 The admin panel shall enforce its own server-side role check.

R8.3 Closed fire incidents shall be moved to a searchable archive automatically.

R8.4 All role changes and access revocations shall be written to the audit log.

4 Use Cases

4.1 Authentication & Access Control

UC-1.1 - User Registration *Actor: Anonymous Visitor* [3]

- This use case begins with an anonymous visitor tapping Register and entering an email address and password.
- This use case ends with the visitor confirming their email and receiving a Registered User role and a JSON Web Token (JWT).

UC-1.2 - User Login *Actor: Registered User, Firefighter, Administrator* [2]

- This use case begins with an account holder tapping Log In and entering their email and password.
- This use case ends with the system validating the credentials and directing the user to their role-appropriate home view with a valid JWT issued.

UC-1.3 - Manage User Roles and Access *Actor: Administrator* [5]

- This use case begins with an administrator opening the user management panel to review pending role requests or search for an account by name or email.
- This use case ends with the administrator approving or rejecting a role request, or revoking an account's access, updating the user's permission accordingly.

UC-1.4 - Reset Password *Actor: Registered User, Firefighter, Administrator* [??]

- This use case begins with an account holder requesting a password reset from the login screen and receiving a time-limited, single-use email link.
- This use case ends with the account holder setting a new password through the link and successfully logging in with the new credentials.

UC-1.5 - Enable Two-Factor Authentication *Actor: Registered User, Firefighter, Administrator* [??]

- This use case begins with an account holder opening account settings and selecting "Enable 2FA".
- This use case ends with the account holder confirming a valid TOTP code from their authenticator app, activating 2FA for future logins.

4.2 Fire Reporting & Verification

UC-2.1 - Fire Report Submission *Actor: Any User (Anonymous Visitor, Registered User, Firefighter or Administrator)* [4]

- This use case begins with any user tapping on the map to report a fire, placing or adjusting the GPS location pin, and entering the required fire details before tapping submit.
- This use case ends with the user receiving a unique reference number as the report is set to Pending status, automatically triggering UC-2.2 (Auto-Verification).

UC-2.2 - Auto-Verification *Actor : System* [4]

- This use case begins with the system detecting a new pending report and checking it against independent nearby reports.

- This use case ends with the system setting the report to Verified and triggering UC-5.1 (Dispatch Proximity Push Alerts), or if no auto-verification condition is met, leaving the report Pending for (UC-2.3 [??]) (Manual Verification).

UC-2.3 - Manual Verification/ Rejection Actor: *Firefighter, Administrator* [4]

- This use case begins with a reviewer selecting a pending fire report to examine its details and attached photos.
- This use case ends with the reviewer verifying the report - triggering UC-3.1 and UC-5.1 - or rejecting it with a mandatory reason and the reporter being notified of the outcome either way.

4.3 AI Fire-Spread Simulation

UC-3.1 - Generate Fire Spread Prediction Actor: *System*

- This use case begins with the system fetching current weather, terrain slope and vegetation data upon a fire report being verified.
- This use case ends with the system publishing a colour-coded spread-probability grid for the 1h/ 3h/ 6h/ 24h horizons to the live map, tagged with the model version used.

UC-3.2 - Re-Run After Containment Line Actor: *System*

- This use case begins with the system receiving newly logged containment-line coordinates for an active verified fire.
- This use case ends with the system re-running the simulation with the containment line as a barrier and publishing the updated prediction to the map within seconds.

UC-3.3 - Auto-Play Fire Spread Simulation Actor: *Firefighter*

- This use case begins with a firefighter enabling auto-play and selecting Run for a verified fire.
- This use case ends with the firefighter having watched the animated playback of the fire's predicted spread from ignition through the simulated period, with the option to pause or resume at any point.

UC-3.4 - Inspect Simulation at a Selected Tick Actor: *Firefighter*

- This use case begins with a firefighter pausing or disabling auto-play and dragging the timeline slider to a specific tick.
- This use case ends with the firefighter viewing the predicted fire extent and burn statistics for that exact tick, without watching the full animation.

4.4 Live Fire Map

UC-4.1 - View Live Fire Map Actor: *Any User* [7]

- This use case begins with any user opening the app to load the map and view verified fire markers, with the option to tap a marker for its status, report time and estimated size.
- This use case ends with the user viewing the map as it auto-refreshes with new data, including the AI spread-prediction overlay and at-risk zones if the user holds a registered role.

4.5 Notification & Alert System

UC-5.1 - Dispatch Proximity Push Alerts *Actor: System*

- This use case begins with the system identifying at-risk registered users via grid-based topic messaging when a fire report changes to Verified.
- This use case ends with the at-risk user tapping the delivered push notification and being taken directly to the fire's location on the map, with the alert logged by timestamp and zone ID.

4.6 Firefighter Tactical Tools

UC-6.1 - View Tactical Dashboard *Actor: Firefighter*

- This use case begins with a firefighter opening the app during an active incident to load the tactical dashboard, showing the live fire map, spread overlay and their own GPS position.
- This use case ends with the firefighter reviewing the AI panel's environmental metrics, high-risk areas and suggested containment locations to inform field decisions.

UC-6.2 - Log Containment Line *Actor: Firefighter*

- This use case begins with a firefighter selecting the "Draw Containment Line" tool and drawing a line on the map, whether online or offline.
- This use case ends with the system saving the line's coordinates and timestamp - locally if offline, syncing once connectivity resumes - and triggering UC-3.2 to re-run the spread prediction with the line as a barrier.

4.7 Offline Map Cache

UC-7.1 - View Map While Offline *Actor: Any User*

- This use case begins with the app detecting a loss of connectivity and displaying an offline indicator.
- This use case ends with the user continuing to view cached map tiles and the most recently received spread prediction, with the app silently refreshing to live data once a stable connection is detected.

4.8 Administration Panel

UC-8.1 - Archive Closed Incident *Actor: System*

- This use case begins with the system detecting that a verified fire incident has been marked closed or contained.
- This use case ends with the system moving the incident's full record into the searchable incident archive.

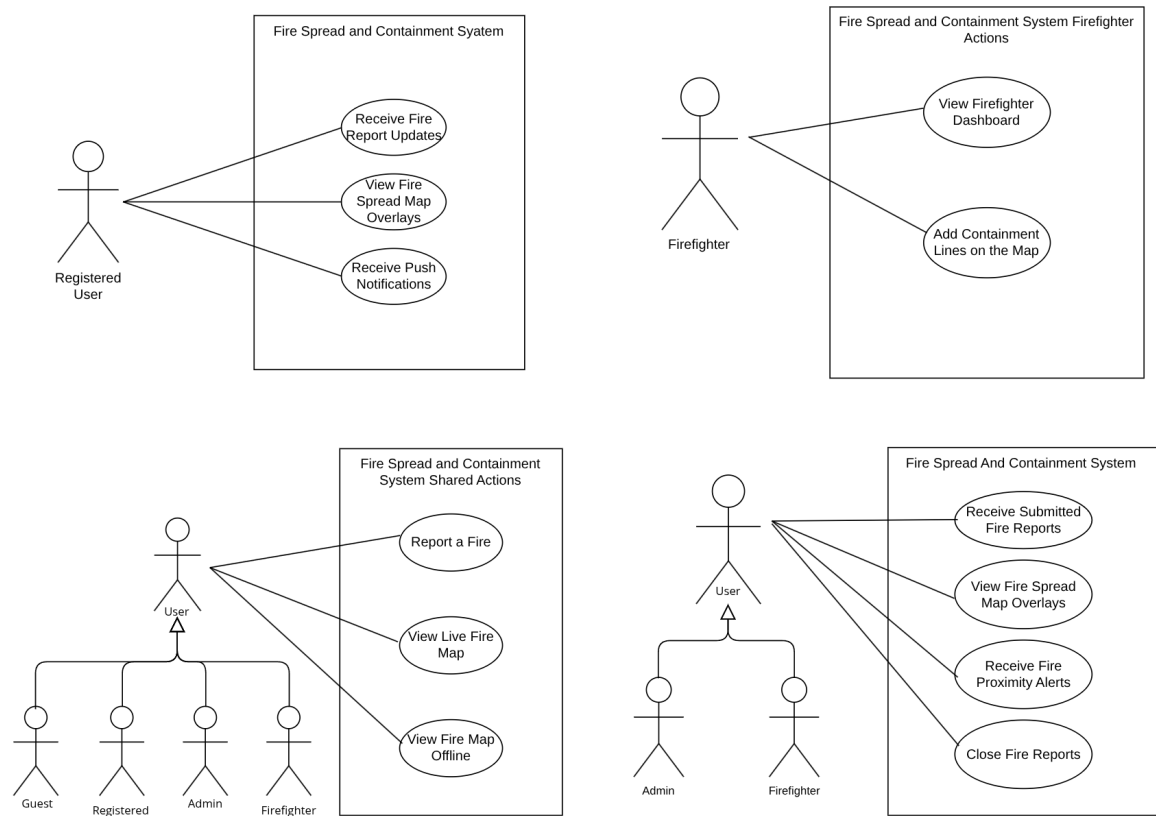


Figure 1: Role Based Access Control

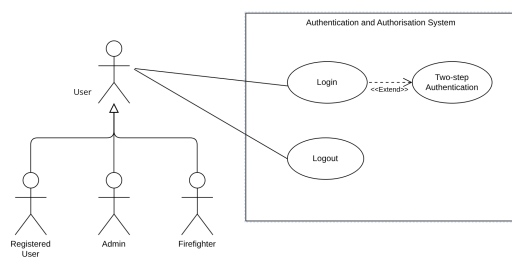


Figure 2: Authentication and Authorisation System

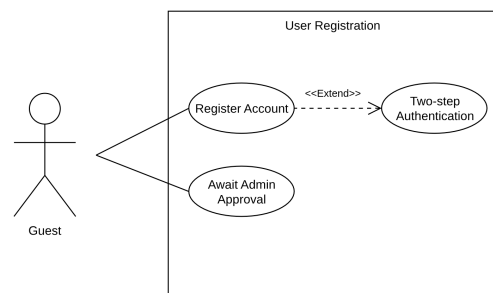


Figure 3: Registration System

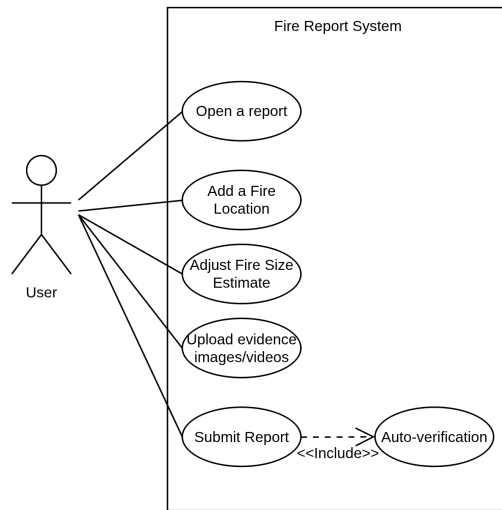


Figure 4: Reporting a Fire System

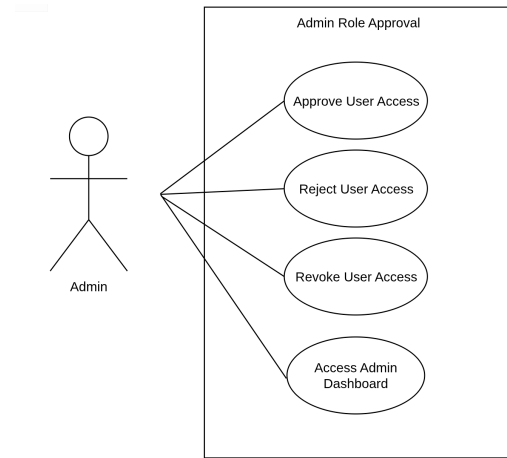


Figure 5: Admin Approval System

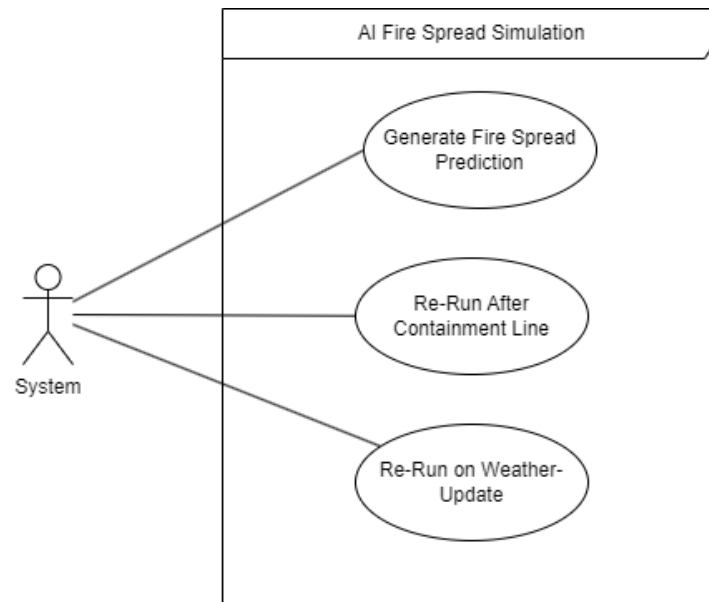


Figure 6: AI Fire Spread simulation

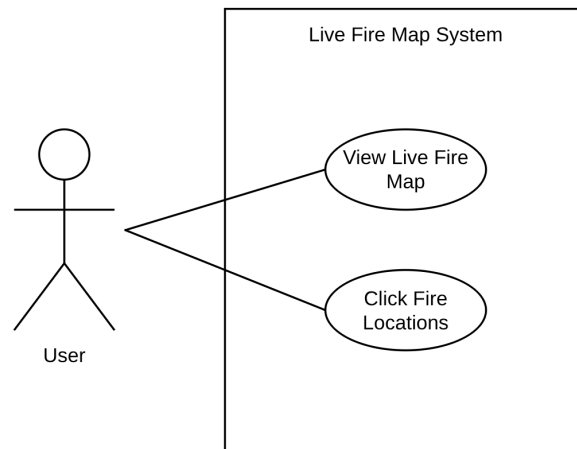


Figure 7: Live Fire Map System

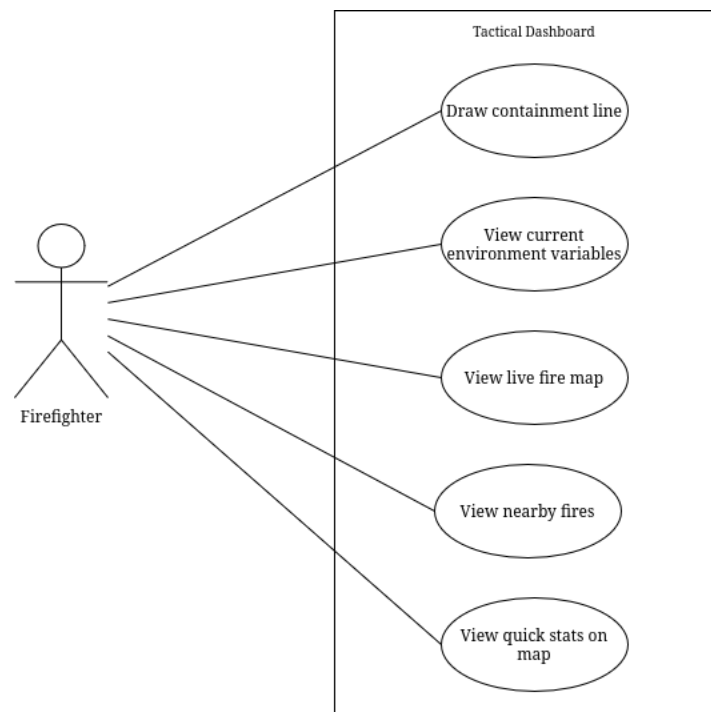


Figure 8: Firefighter Tactical Dashboard System

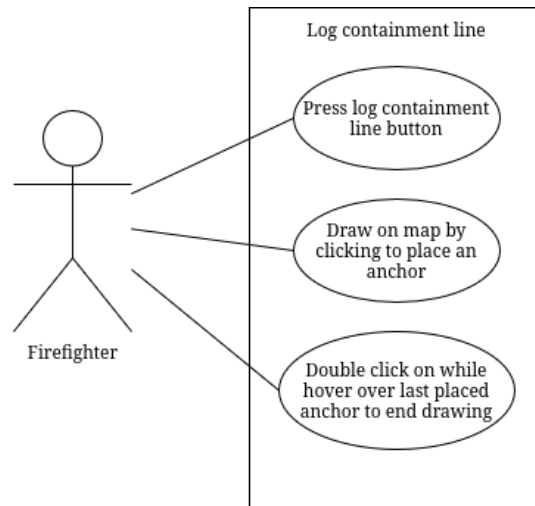


Figure 9: Log Containment Line

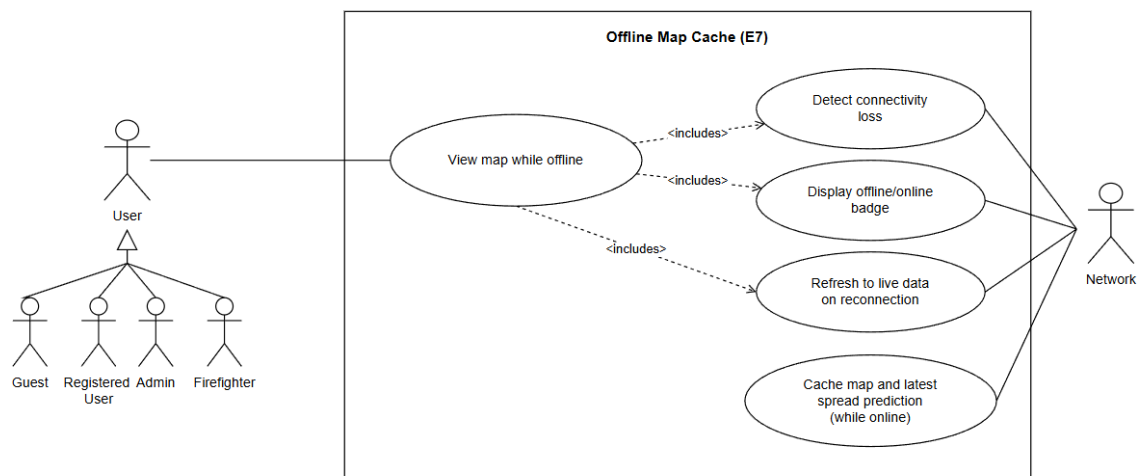


Figure 10: Offline Map Cache System

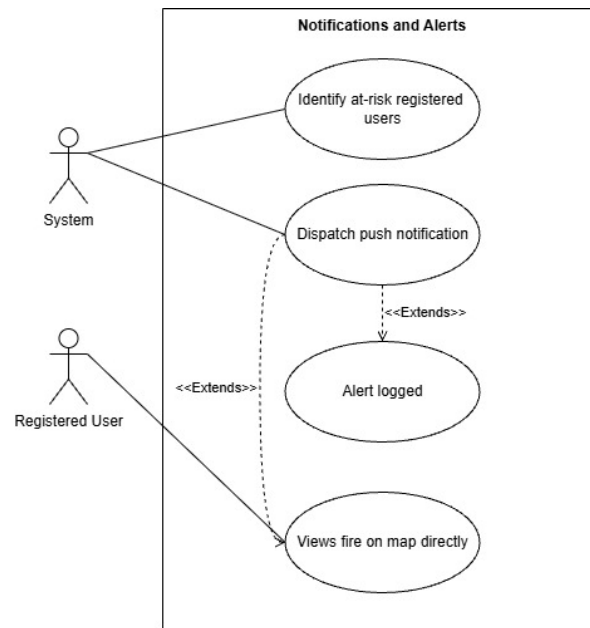


Figure 11: Notification and Alert System

5 Non-Functional Requirements

5.1 Performance & Simulation Latency

- NFR-01.1 The system shall generate and render a fire spread prediction grid for 1h, 3h, 6h, and 24h horizons within 3.0 seconds for at least 95% of requests under normal load.
- NFR-01.2 The system shall deliver initial live simulation results to the client within 5.0 seconds under peak load.
- NFR-01.3 The client map shall redraw GPU-accelerated overlay polygons and probability layers within 16 milliseconds per frame (at least 60 FPS).
- NFR-01.4 The system shall return cached weather and terrain data in under 50 milliseconds without calling external APIs.

5.2 Scalability & Workload Capacity

- NFR-02.1 The system shall support at least 1,000 concurrent active users submitting queries and reports with an HTTP error rate below 1.0%.
- NFR-02.2 The backend shall process concurrent fire report submissions with a p95 response time under 400 milliseconds during a 5-minute test with 200 virtual users.

5.3 Security & Access Control

- NFR-03.1 The system shall enforce Role-Based Access Control so that all admin and firefighter endpoints reject unauthorized requests with HTTP 403.
- NFR-03.2 The system shall encrypt all sensitive user data at rest with AES-256 and enforce TLS 1.3 on all public network traffic.
- NFR-03.3 The login service shall rate-limit repeated failures: an HTTP 429 delay (30 seconds) after 5 failed attempts, and an HTTP 423 lockout after 10.
- NFR-03.4 The system shall authenticate all incoming data from external sources (NASA FIRMS, weather APIs) using signed tokens or pre-shared keys before ingesting it.

5.4 Availability & Offline Resilience

- NFR-04.1 The web and mobile platforms shall maintain at least 99.9% uptime over any 30-day period.
- NFR-04.2 The client shall remain usable when offline, rendering all previously cached map tiles and incident markers from local storage.
- NFR-04.3 The client shall keep offline-logged containment lines and queued reports in local storage, syncing them automatically within 30 seconds of reconnecting.

5.5 Reliability & Fault Recovery

- NFR-05.1 The system shall record all containment line changes and role updates to the audit log with millisecond-precision timestamps.
- NFR-05.2 The simulation shall run to completion without crashing when up to 25% of environmental data is missing, falling back to a degraded model.

5.6 Usability & Portability

NFR-07.1 The interface shall meet WCAG 2.1 AA, scoring 95 or above on automated accessibility audits.

NFR-07.2 The app shall render correctly on mobile, tablet, and desktop, with tap targets at least 48px by 48px.

NFR-07.3 Components shall run in standard containers, deploying identically across local and production environments without manual setup.

6 Domain Model

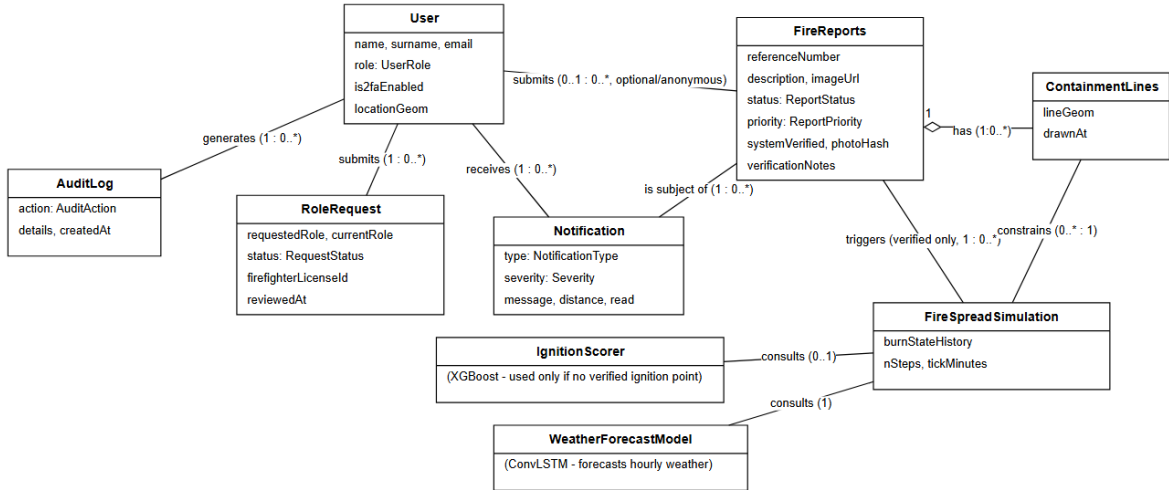


Figure 12: AI-Powered Fire Spread Prediction and Containment System Domain Model

7 Change Log

Date	Version	Change
May 2026	1.0	Initial version
July 2026	2.0	Use Cases, NFR
August 2026	3.0	Use Cases, NFR